Md Ahsanul Haque, Ph.D.
MAT-120

Quiz-1

Date: February 22, 2026

Name

Section

Instructions:

- Write your full name and section number.
- This quiz contains 2 problems. It is **YOUR responsibility** to check that your copy has all 2 problems.
- Make sure to read the instructions for each problem carefully.
- Use correct mathematical notation.
- Write cohesively and follow a logical sequence of steps.
- Check that your answer can be read and understood by the grader.

1. (a) Let, $f(x) = \sqrt{x+1}$ and $g(x) = \ln(3-x)$.
   Find the composite function $(g \circ f)(x)$ and determine its domain. Express your answer using interval notation.

   (b) Consider the function $h(x) = \frac{2e^x - 1}{e^x + 5}$. Find the inverse function, $h^{-1}(x)$.

   (c) If the graph of $p(x) = x^2$ is shifted 3 units to the right, vertically stretched by a factor of 2, and then reflected across the x-axis, write the equation for the new function $q(x)$.

2. Evaluate the following limits. If a limit does not exist, state DNE and explain why.

   (a)
   $$\lim_{x \to 4} \frac{\frac{1}{x} - \frac{1}{4}}{x - 4}$$

   (b)
   $$\lim_{h \to 0} \frac{\sqrt{25+h} - 5}{h}$$

   (c) Let, $k(x) = \frac{5x^3 - 2x + 1}{\sqrt{4x^6 + 9}}$.
   $$\lim_{x \to \infty} k(x)$$